

EKIP Project Handbook

Enterprise Knowledge Intelligence Platform

EKIP is a multi-tenant enterprise knowledge platform that solves a common problem: employees waste hours searching across disconnected systems such as PDFs, Word documents, Excel sheets, and PowerPoint files. EKIP gives users one intelligent search interface that understands context, respects access permissions, and returns answers with source citations.

RBAC Enforcement Policy

Role-Based Access Control in EKIP happens BEFORE retrieval, not after. Unauthorized documents must never reach the language model, even to be filtered out later. This is enforced structurally: every vector stored in Qdrant carries organization, workspace, department, and allowed role metadata at write time. Search queries apply these as hard filters, so unauthorized content is excluded before the retrieval step even runs.

Hybrid Retrieval

EKIP combines dense retrieval using Sentence Transformer embeddings with sparse retrieval using BM25. Results from both approaches are combined through Reciprocal Rank Fusion, then passed through a cross-encoder reranker before being sent to the Ollama language model for final answer generation. Citations are extracted by mapping the generated answer back to the exact source chunks that informed it.